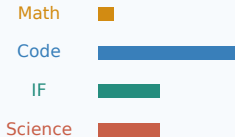


Spend more samples on less precise task updates

1 Measure variability

Ordinary dense-loss gradients

Apply optimizer scaling D

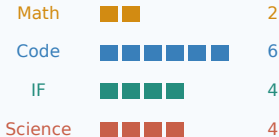


Measured noise e_i

2 Allocate micro-batches

$$m_i \propto w_i \sqrt{e_i}$$

Example: 16 micro-batches



More noise, more samples

3 Keep task weights fixed

Average within each task

$$A = \frac{1}{4} \bar{g}_{\text{Math}} + \frac{1}{4} \bar{g}_{\text{Code}} + \frac{1}{4} \bar{g}_{\text{IF}} + \frac{1}{4} \bar{g}_{\text{Science}}$$

Different sample counts
Same intended task balance

Then take the AdamW step

Reuse this step's statistics to allocate the next step

Cost-aware mode also uses measured time per task